

Constructing investment strategy portfolios by combination genetic algorithms

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Abstract

The classical portfolio problem is a **problem of distributing capital to a set of securities**. By generalizing the set of securities to a set of investment strategies (or security-rule pairs), this study proposes an investment strategy portfolio problem, which **becomes a problem of distributing capital to a set of investment strategies**.

Since the investment strategy portfolio problem can be formulated as a combination optimization problem, a new combination genetic algorithm is proposed for solving the new investment strategy portfolio problem. Experimental results show that the idea of investment strategy portfolios is feasible and the combination genetic algorithm is effective on the investment strategy portfolio problem.

Classical portfolio optimization problem

$$\begin{aligned} \max \quad & \alpha \sum_i x_i r_i - (1 - \alpha) \sum_{ij} x_i x_j \sigma_{ij} \\ \text{subject to} \quad & \sum_{i=0}^n x_i = 1, 0 \leq x_i \leq 1, x_i \in R, \end{aligned} \tag{1}$$

α : a real constant between 0 & 1

x_i : the proportion (0~1) of capital invested in instrument i

r_i : the expected return rate of instrument i

Σ_{ij}^2 : the covariance of r_i and r_j

goal : find the optimal portfolio with max return and min risk

It can be viewed **a special case of the new investment strategy portfolio problem with buy-and-hold as the only trading rule.**

Investment strategy portfolio problem

- An investment strategy is a security-rule pair.
- A trading rule consists of a **buy condition** and a **sell condition**.
- The buy/sell conditions can be base on various factors.

Variants of the investment strategy portfolio problem :

- 1 、 multiple securities with a single trading rule
- 2 、 a single security with multiple trading rules(Experiment 1)
- 3 、 multiple securities with multiple trading rules(Experiment 2)

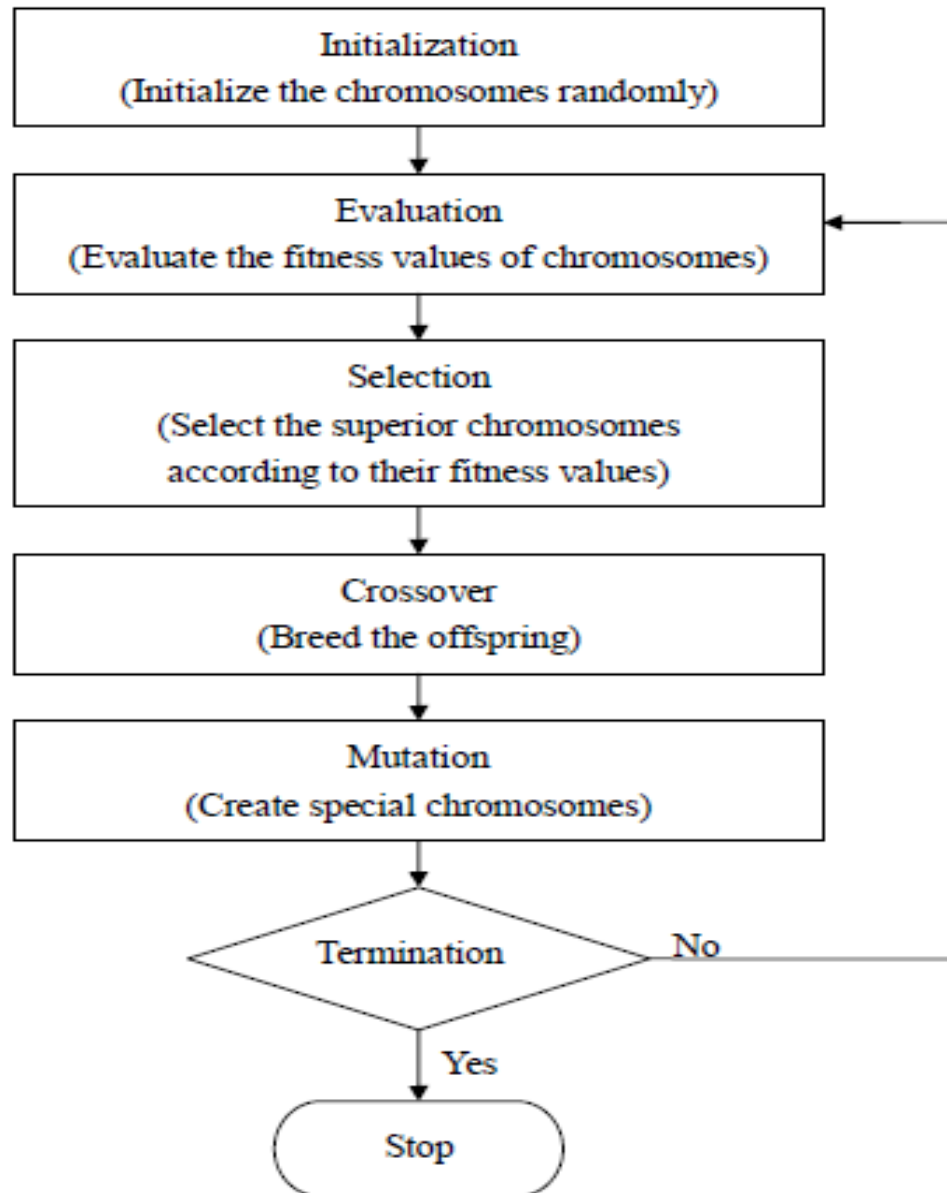
Combination genetic algorithms

Modified the original problem as below :

$$\begin{aligned} \max \quad & \alpha \sum_i \frac{w_i}{m} \cdot r_i - (1 - \alpha) \sum_{ij} \frac{w_i}{m} \cdot \frac{w_j}{m} \cdot \sigma_{ij} \\ \text{subject to} \quad & \sum_{i=0}^n w_i = m, \quad 0 \leq w_i \leq m, \quad w_i \in \mathbb{Z}. \end{aligned}$$

- m is a large positive integer
- w_i/m approximates the x_i in (1), $0 \leq w_i \leq m, w_i \in \mathbb{Z}$
- Size of search space is $C(n+m, m)$

Genetic algorithm



Combination encoding

$\sum_{i=0}^n w_i = m$ have a 1-1 correspondence with $(n+m)$ -bit strings containing exactly m 1s.

We use 0s as the delimiters for w_i 's and the consecutive 1s as unary numbers for w_i 's

Ex : $n = 2, m = 5$

Chromosome : 1110110, length = $(5+2) = 7$

Represents that $w_0=3, w_1=2, w_2=0$

Combination crossover

- Propose a shrinking crossover(SX)
- Two-point crossover
- Can't violate the m 1s constraint!!

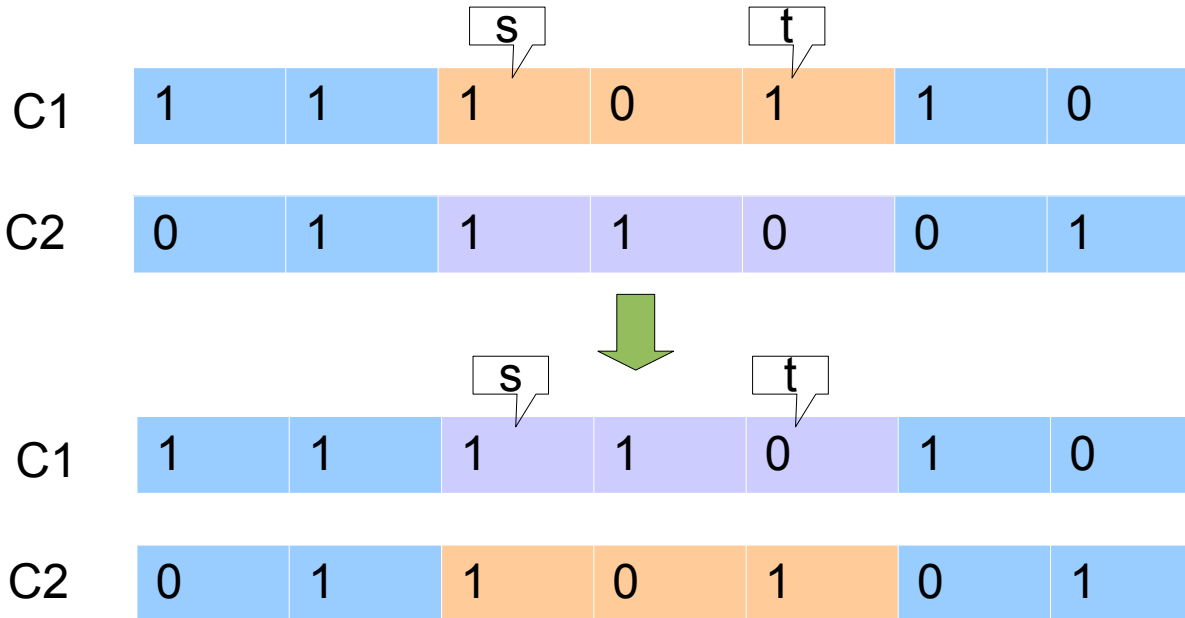
Select two crossover points s and t

C1	1	1	1	0	1	1	0
C2	0	1	1	1	0	0	1

Because C1 & C2 don't have equal number of 1s between s and t, shrink t

C1	1	1	1	0	1	1	0
C2	0	1	1	1	0	0	1

Crossover the elements between s and t



procedure SX (C1, C2)

 select two crossover points s & t

until (C1 & C2 have equal number of 1s between s & t)

 t = t-1

enduntil

for (i = s to t)

 Temp = C1 [i]

 C1 [i] = C2 [i]

 C2 [i] = Temp

endfor

endprocedure

Fig. 2. Algorithm of the shrinking crossover (SX).

Experiment 1

A single security with multiple trading rules

- Dow Jones Industrial Average, IBM
- Data period : 07/2001 ~12/2005
- Trading rules(buy and sell) : $10 \times 10 = 100$
- Security-rule pairs : $1 \times 100 = 100$

No.	Indicator	Buy signal	Sell signal
1	CCI	CCI $\nearrow$ -100	CCI $\searrow$ 100
2	MOM	MOM $\nearrow$ 0	MOM $\searrow$ 0
3	RSI	RSI line $\nearrow$ 30	RSI line $\searrow$ 70
4	DMI	+DI $\nearrow$ -DI	+DI $\searrow$ -DI
5	MACD	DIF $\nearrow$ MACD or 0	DIF $\searrow$ MACD or 0
6	KD	K $\nearrow$ D & D < 20	K $\searrow$ D & D > 20
7	WMS %R	WMS %R $\searrow$ 80	WMS %R $\nearrow$ 20
8	BIAS	BIAS $\nearrow$ -4.5%	BIAS $\searrow$ 5%
9	Qstick	Short Qstick $\nearrow$ Long Qstick or 0	Short Qstick $\searrow$ Long Qstick or 0
10	PSY	PSY $\nearrow$ 25%	PSY $\searrow$ 75%

- The CGA parameters

- Training 9 months, testing 3 months, shifting 3 months

Window 1:

Training	Testing
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Window 2:

Training	Testing
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Window k :

⋮

ℓ months	s month
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- Only the best 10% performing pairs are used to construct the investment strategies. ($n = 100 * 10\% = 10$)
- Total capital is divided into $m = 50$ equal portions.
- The length of chromosome = $(50+10) = 60$ bits
- Population size is 200.
- Crossover SX, rate = 100%
- Mutation rate : 0.5%
- Terminal condition : 500 generations

Fitness function :

$$f(w_0, w_1, \dots, w_n) = \prod_{t=1}^{\ell} (1 + r_p(t))$$

where $r_p(t) = \sum_{i=0}^n w_i r_i(t)$ is the return of the portfolio during month t in the training period.

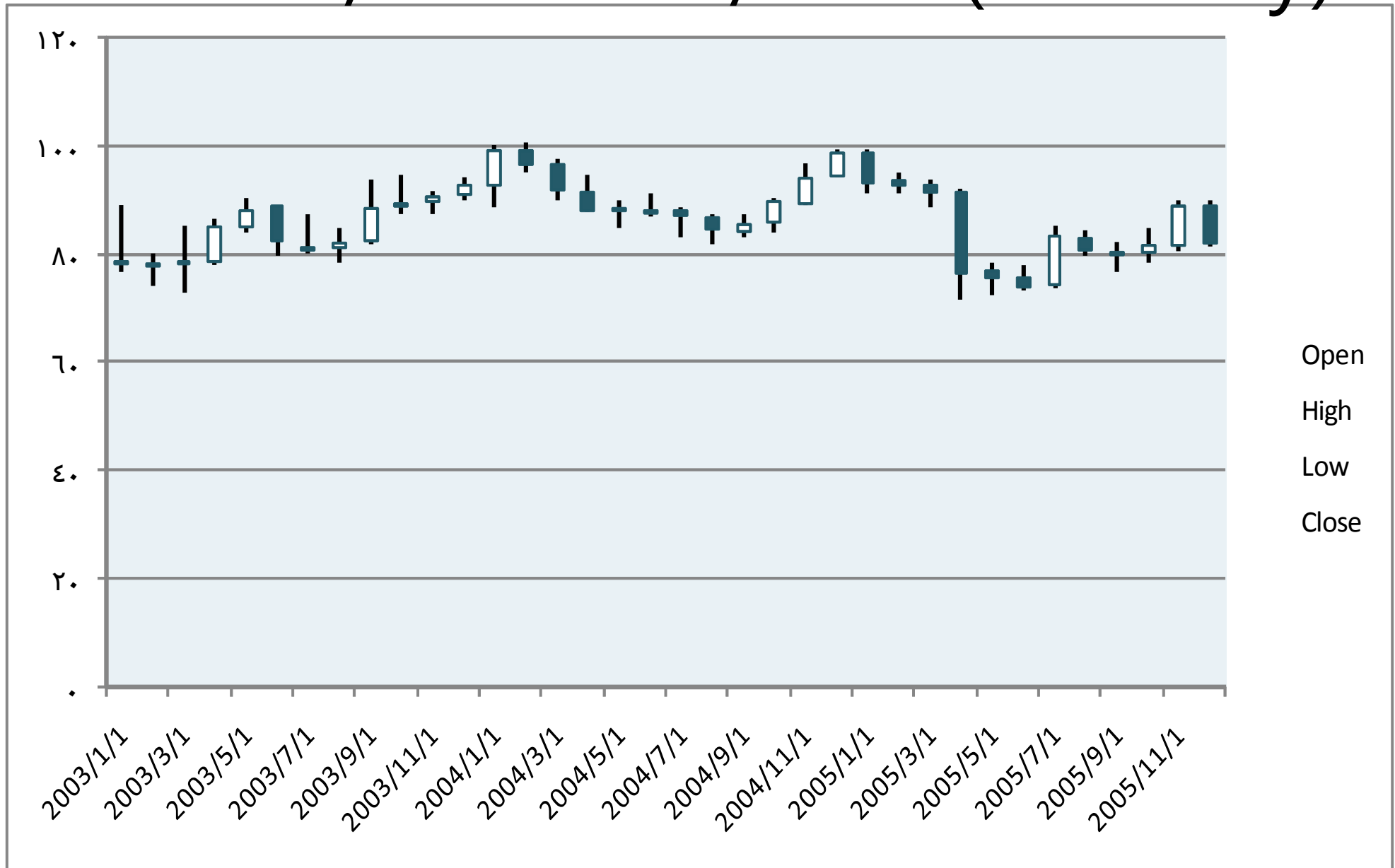
Result(run 5 times with different initial random populations) :

Table 2

Average portfolio returns of CGA on the single security problem

Testing period	Run1	Run2	Run3	Run4	Run5	Average
2003/01/01–03/31	−0.73	−1.62	−1.70	−1.72	−1.89	−1.53
2003/04/01–06/30	3.17	2.99	3.50	2.84	3.20	3.14
2003/07/01–09/30	−0.31	−1.40	−1.87	−0.97	−0.50	−1.01
2003/10/01–12/31	−1.02	−0.35	−0.78	−0.72	−0.24	−0.62
2004/01/01–03/31	−0.81	−1.15	−1.65	−1.16	−1.84	−1.32
2004/04/01–06/30	−4.93	−3.60	−3.49	−1.68	−1.97	−3.13
2004/07/01–09/30	−0.40	−0.24	−0.72	−1.31	−0.86	−0.71
2004/10/01–12/31	0.02	0.00	0.00	0.11	0.04	0.04
2005/01/01–03/31	−3.95	−4.25	−3.24	−4.75	−5.65	−4.37
2005/04/01–06/30	−1.08	−0.54	−1.46	−0.88	−2.53	−1.30
2005/07/01–09/30	0.60	1.01	1.01	0.49	0.97	0.81
2005/10/01–12/31	−0.31	−0.64	0.84	0.53	0.89	0.26
Accumulated	−9.54	−9.56	−9.34	−9.01	−10.14	−9.50

IBM historical price from 01/2003~12/2005(monthly)



Result(compare between CGA and uniform allocation) :

Table 3

Comparisons between CGA and uniform allocation on the single security problem

Testing period	CGA	Uniform
2003/01/01–03/31	−1.53	−1.62
2003/04/01–06/30	3.14	−1.16
2003/07/01–09/30	−1.01	0.72
2003/10/01–12/31	−0.62	−0.91
2003	−0.09	−2.95
2004/01/01–03/31	−1.32	−1.14
2004/04/01–06/30	−3.13	−2.54
2004/07/01–09/30	−0.71	−1.43
2004/10/01–12/31	0.04	3.26
2004	−5.05	−1.93
2005/01/01–03/31	−4.37	−3.38
2005/04/01–06/30	−1.3	−4.36
2005/07/01–09/30	0.81	0.66
2005/10/01–12/31	0.26	0.06
2005	−4.59	−6.93
Accumulated	−9.50	−11.42

- Paired t test between CGA and uniform allocation has a p value = 0.001958 < 0.01
- CGA is significantly better than that of uniform allocation at 99 % confidence level.

Experiment2

Multiple securities with multiple trading rules

- 10 stocks

DJIA : IBM, BA, MMM, CAT, MO, AIG, PG, XOM, JNJ, UTX

- Data period : 07/2001 ~12/2005

- Security-rule pairs : $10 \times 100 = 1000$

- Use the best 10% performing pairs : $n = 1000 \times 10\% = 100$

- Total capital is divided into $m = 50$ equal portions.

- GA search space : $(n=100, m = 50)$

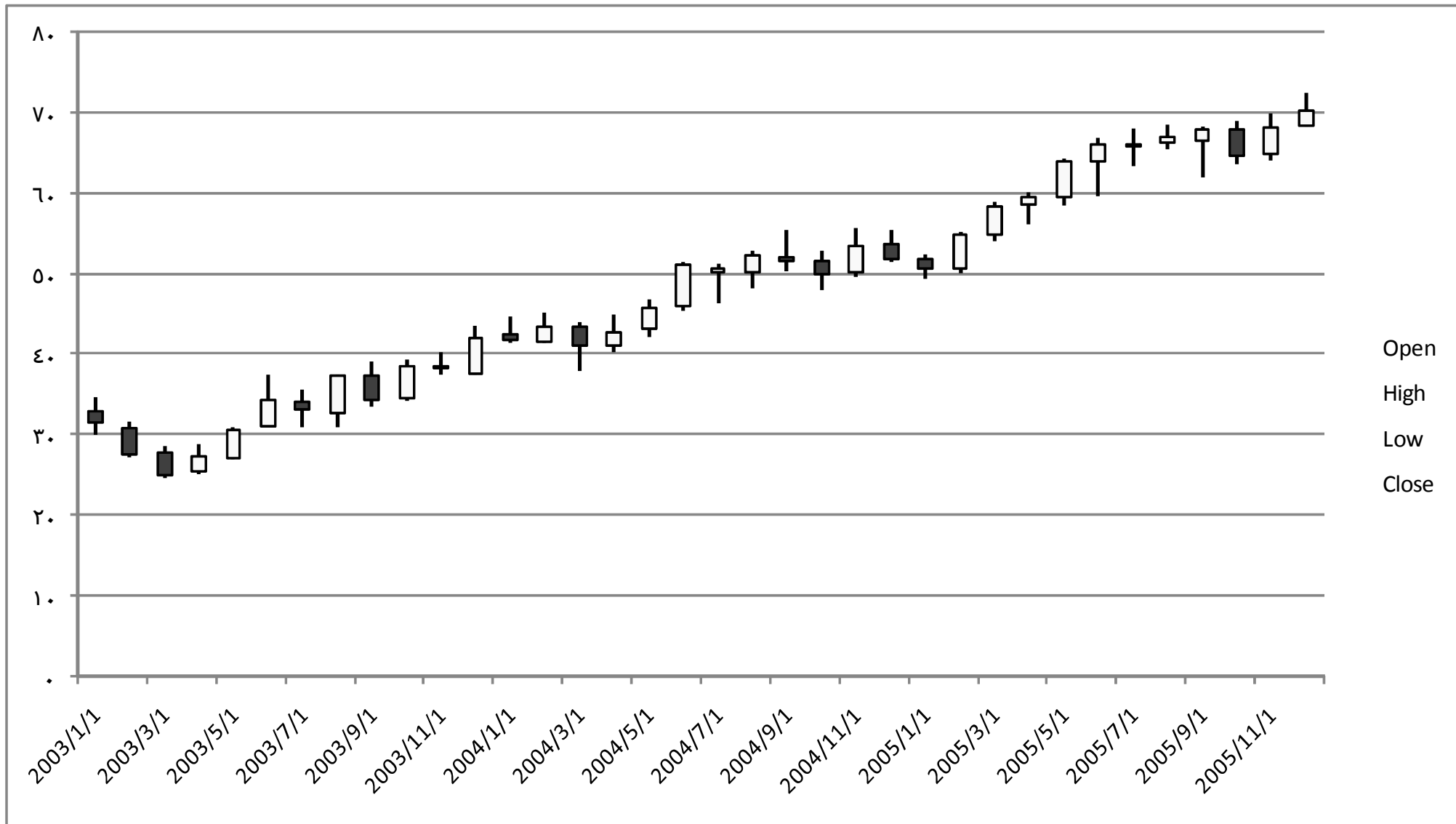
$$C_{50}^{150} = \frac{150!}{50!100!} \approx 2.01 \times 10^{40}$$

Table 4

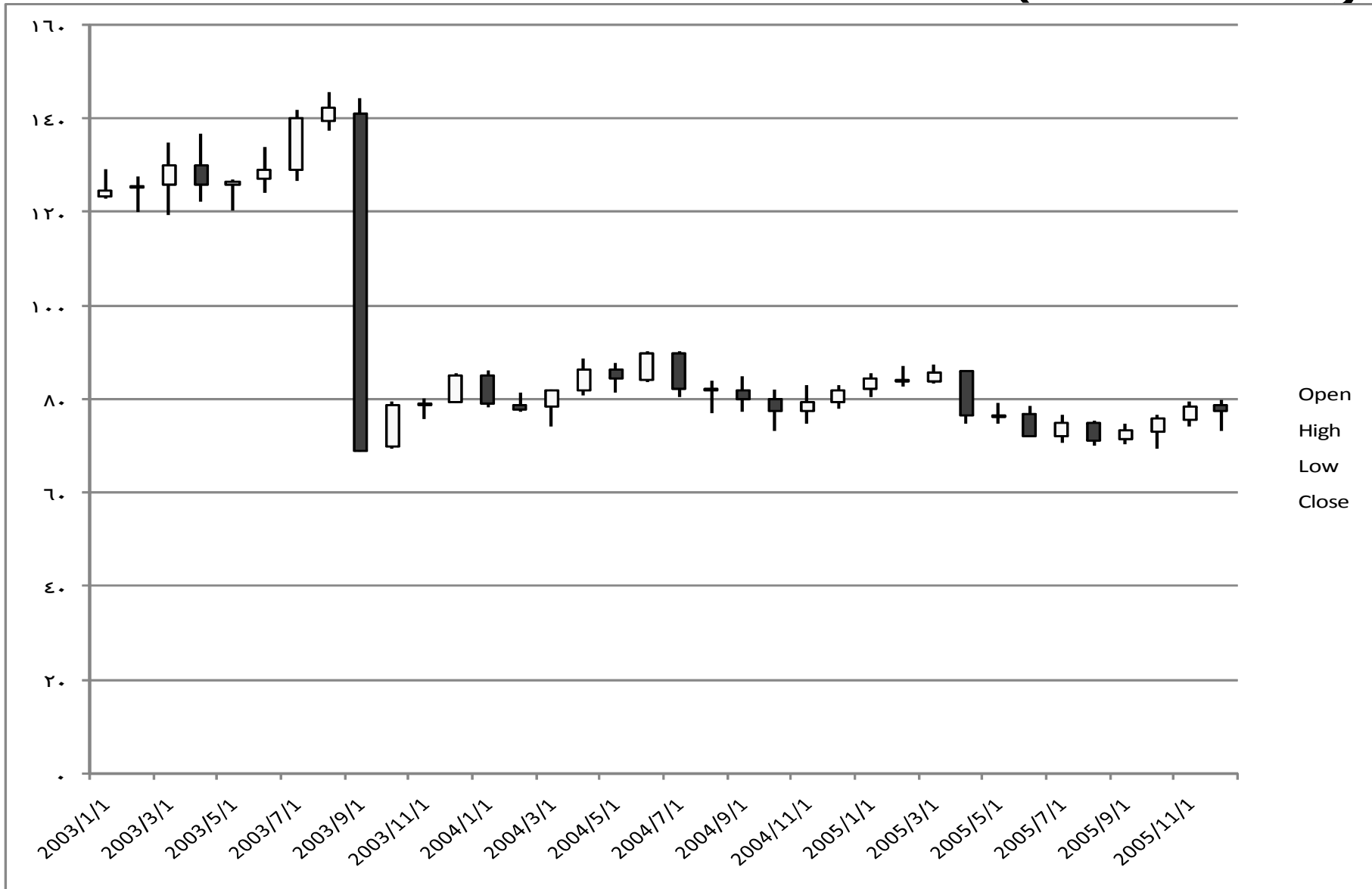
Top 10 priced stocks of DJIA

No.	Symbol	Company name
1	IBM	International Business Machines Corp.
2	BA	Boeing Co.
3	MMM	3M Co.
4	CAT	Caterpillar Inc.
5	MO	Altria Group Inc.
6	AIG	American International Group, Inc.
7	PG	Procter & Gamble Co.
8	XOM	Exxon Mobil Corp.
9	JNJ	Johnson & Johnson DC
10	UTX	United Technologies Corp.

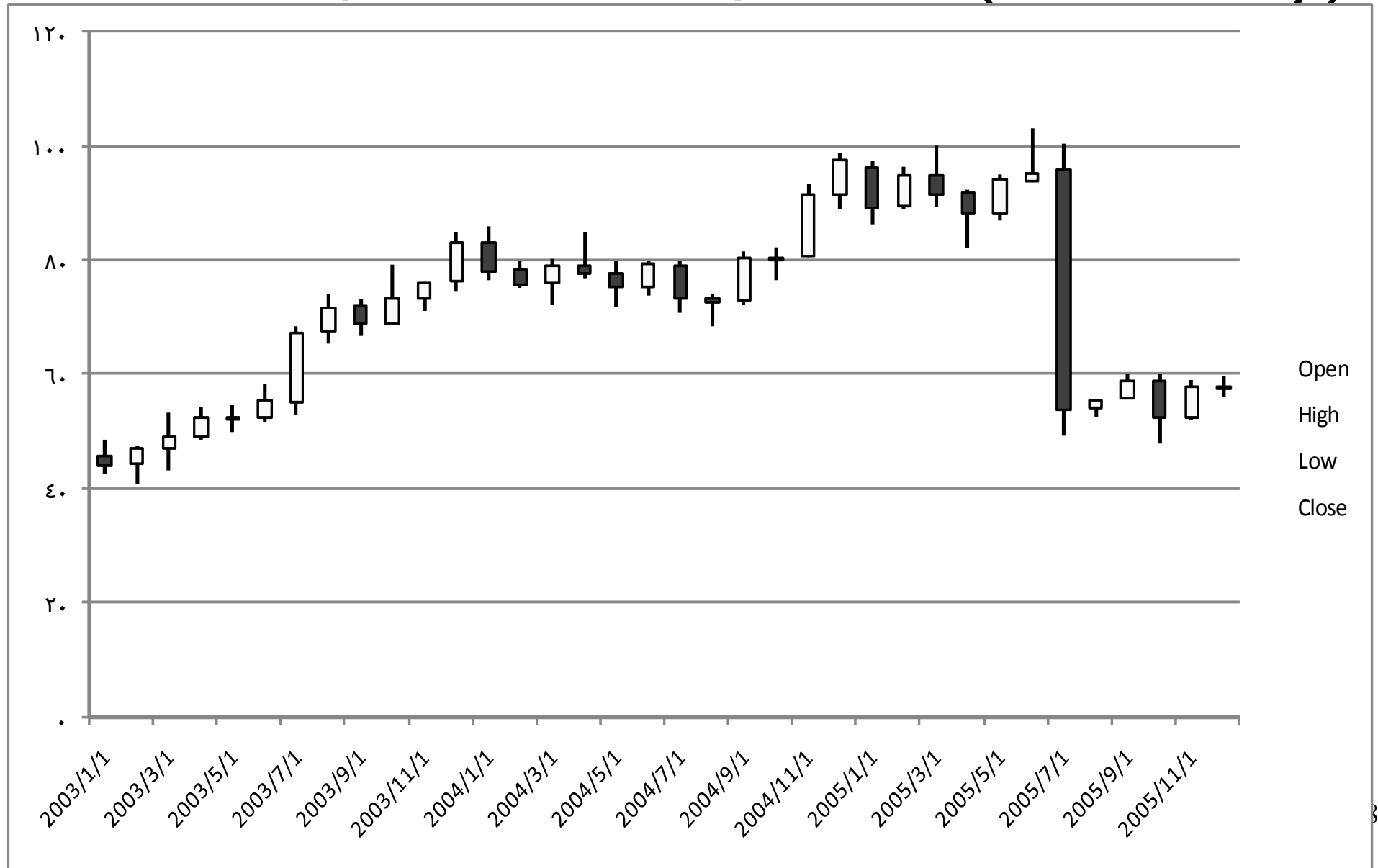
Boeing Co. historical price from 01/2003~12/2005(monthly)



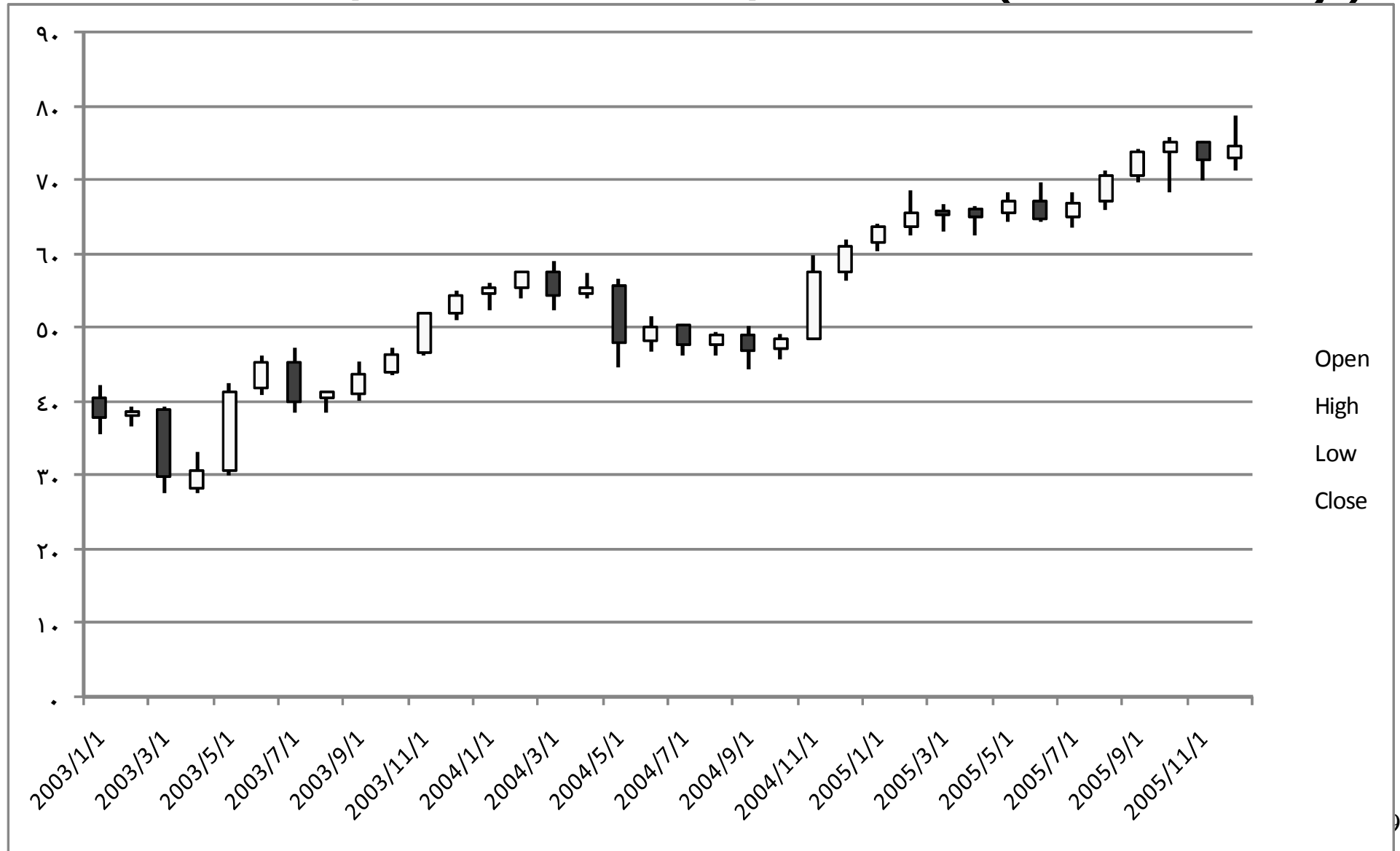
3M Co. historical price from 01/2003~12/2005(monthly)



Caterpillar Inc. historical price from 01/2003~12/2005(monthly)

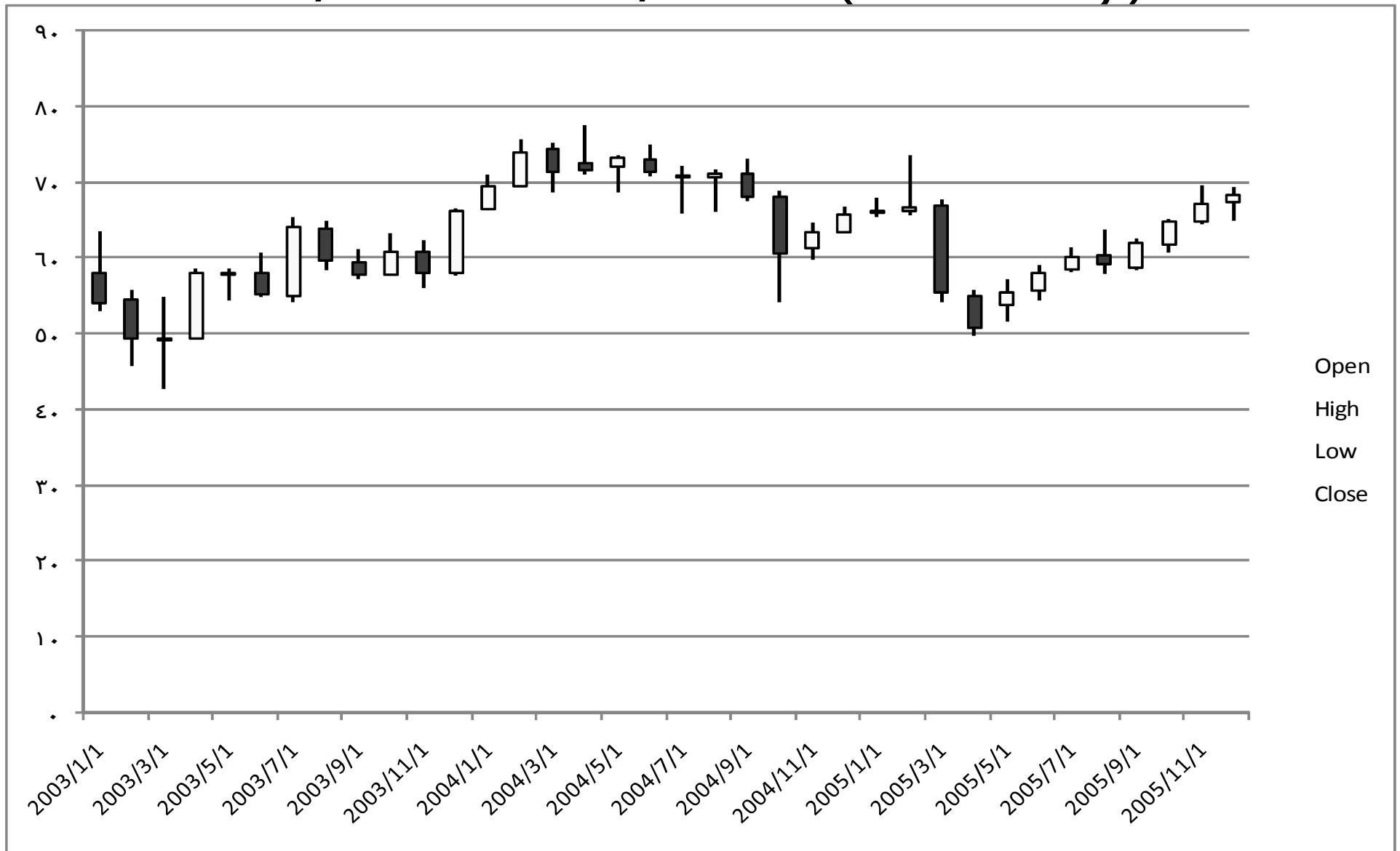


Altria Group Inc. historical price from 01/2003~12/2005(monthly)

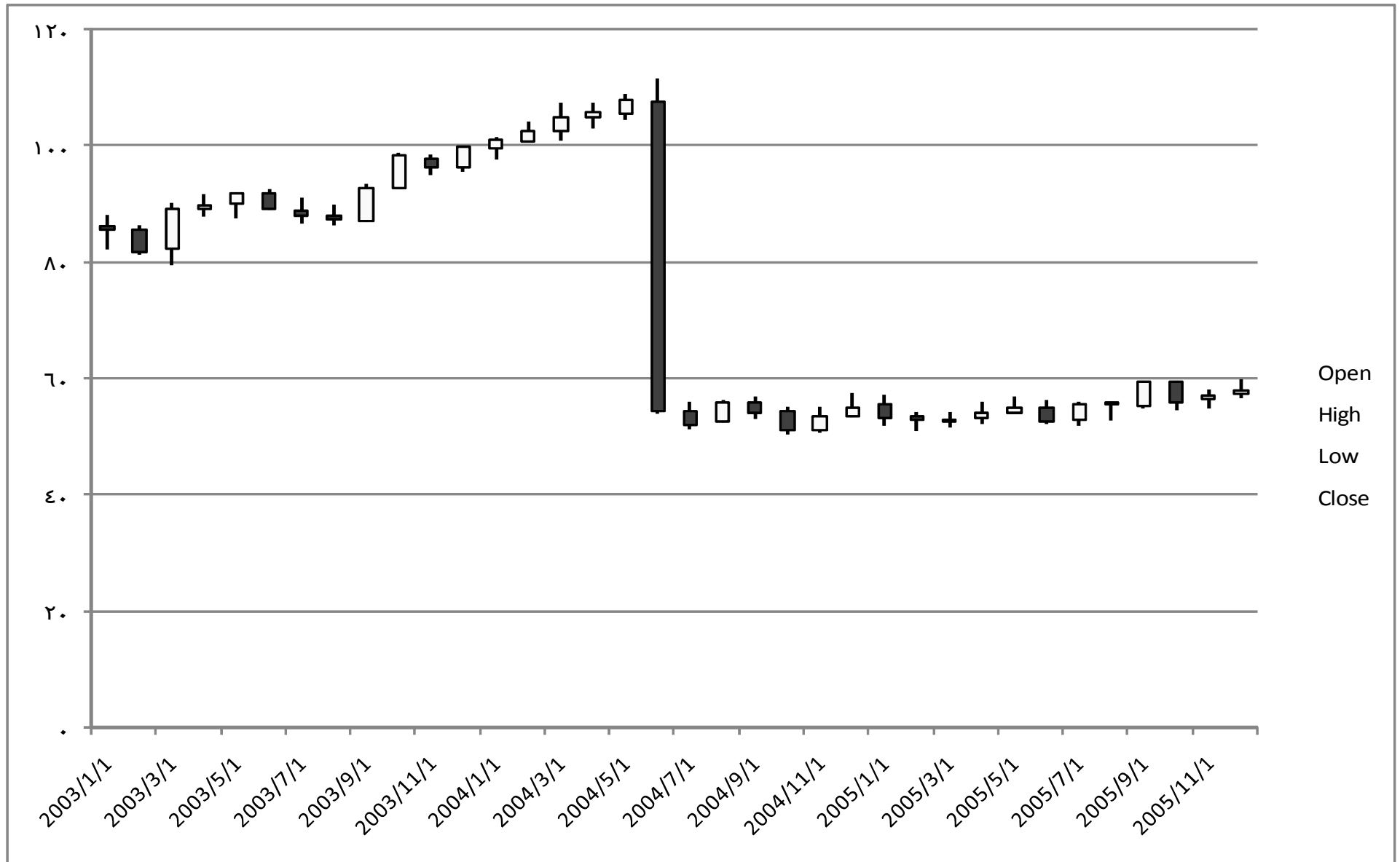


American International Group, Inc.

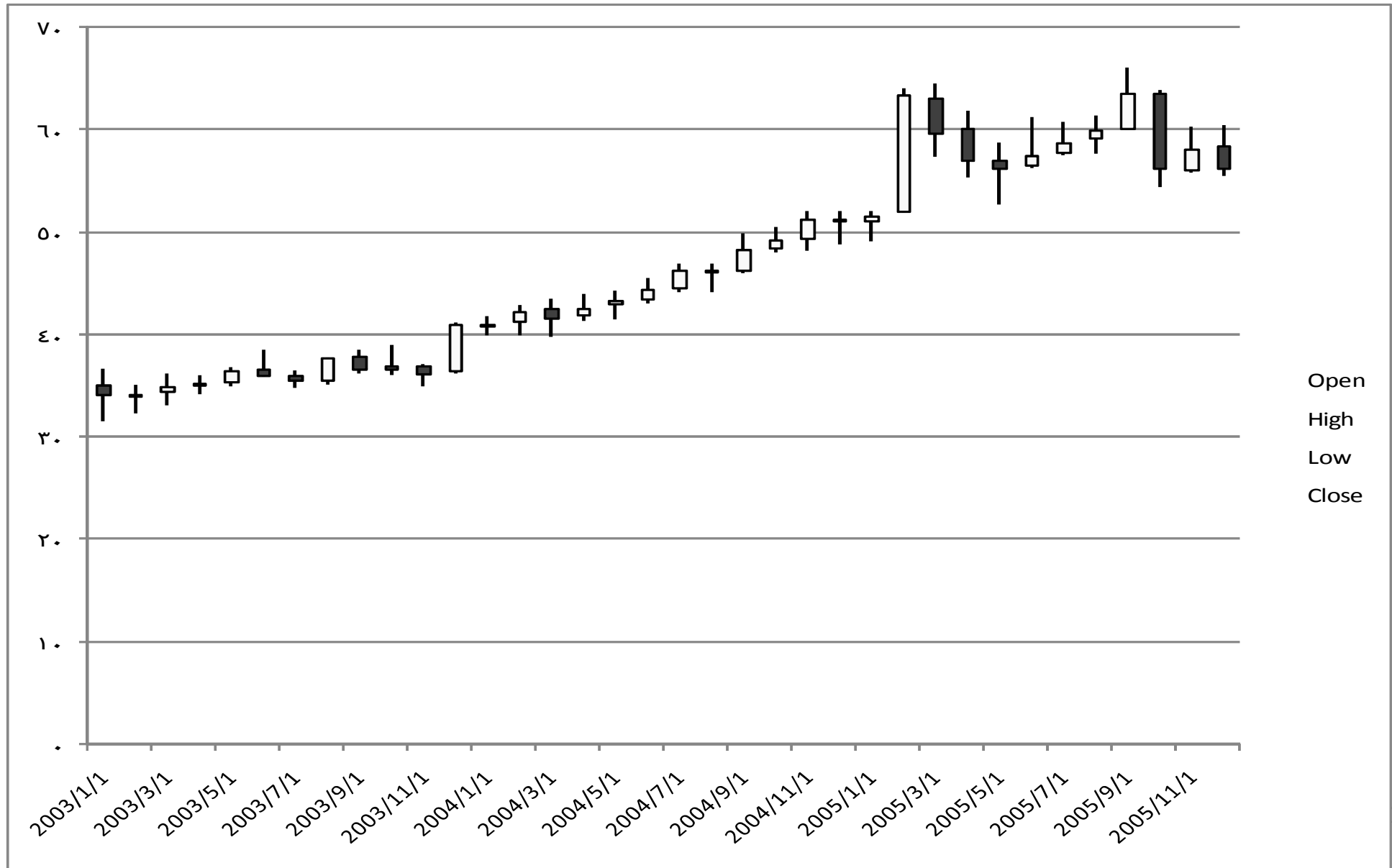
historical price from 01/2003~12/2005(monthly)



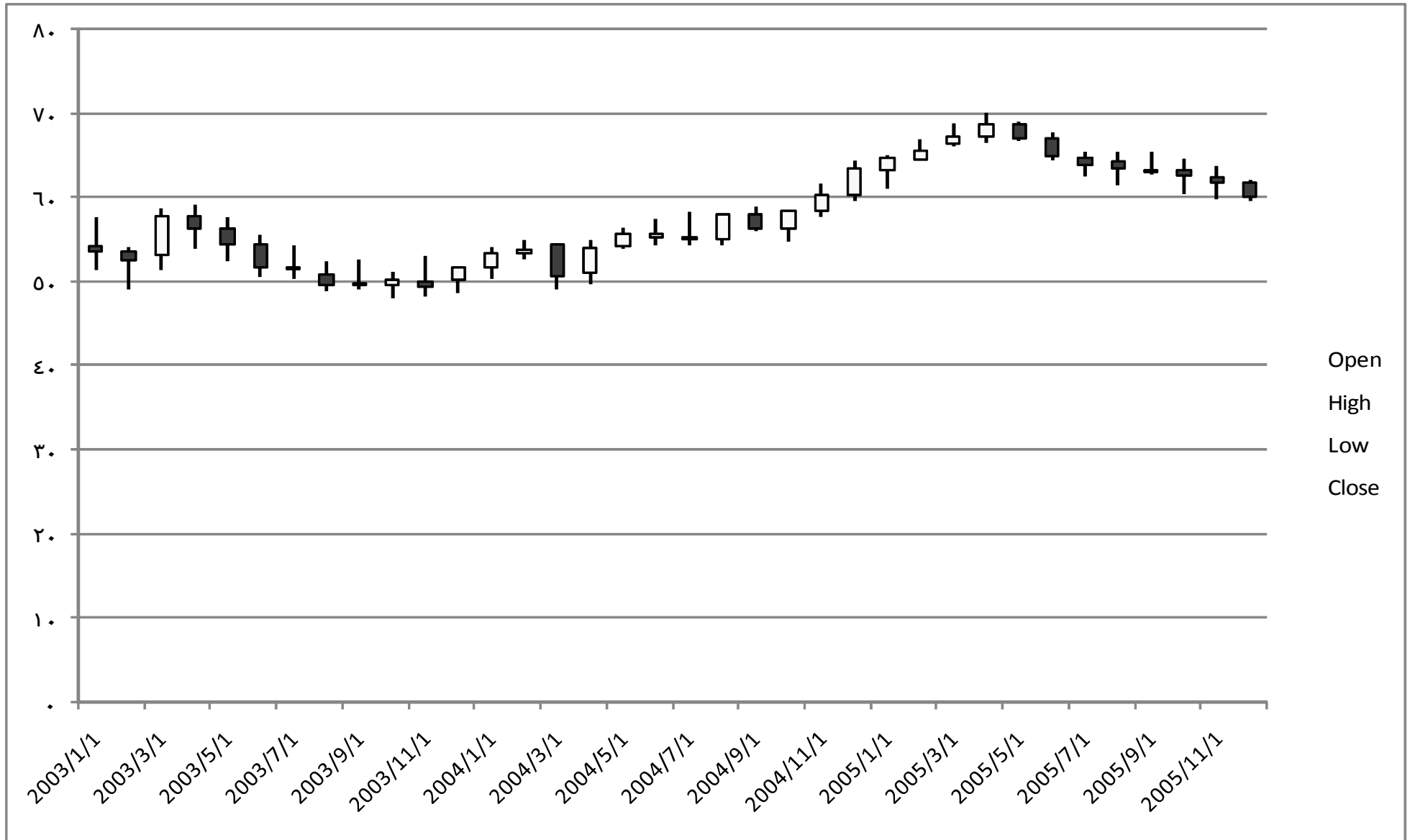
Procter & Gamble, Co. historical price from 01/2003~12/2005(monthly)



Exxon Mobil Corp. historical price from 01/2003~12/2005(monthly)



Johnson & Johnson DC historical price from 01/2003~12/2005(monthly)



United Technologies Corp. historical price from 01/2003~12/2005(monthly)

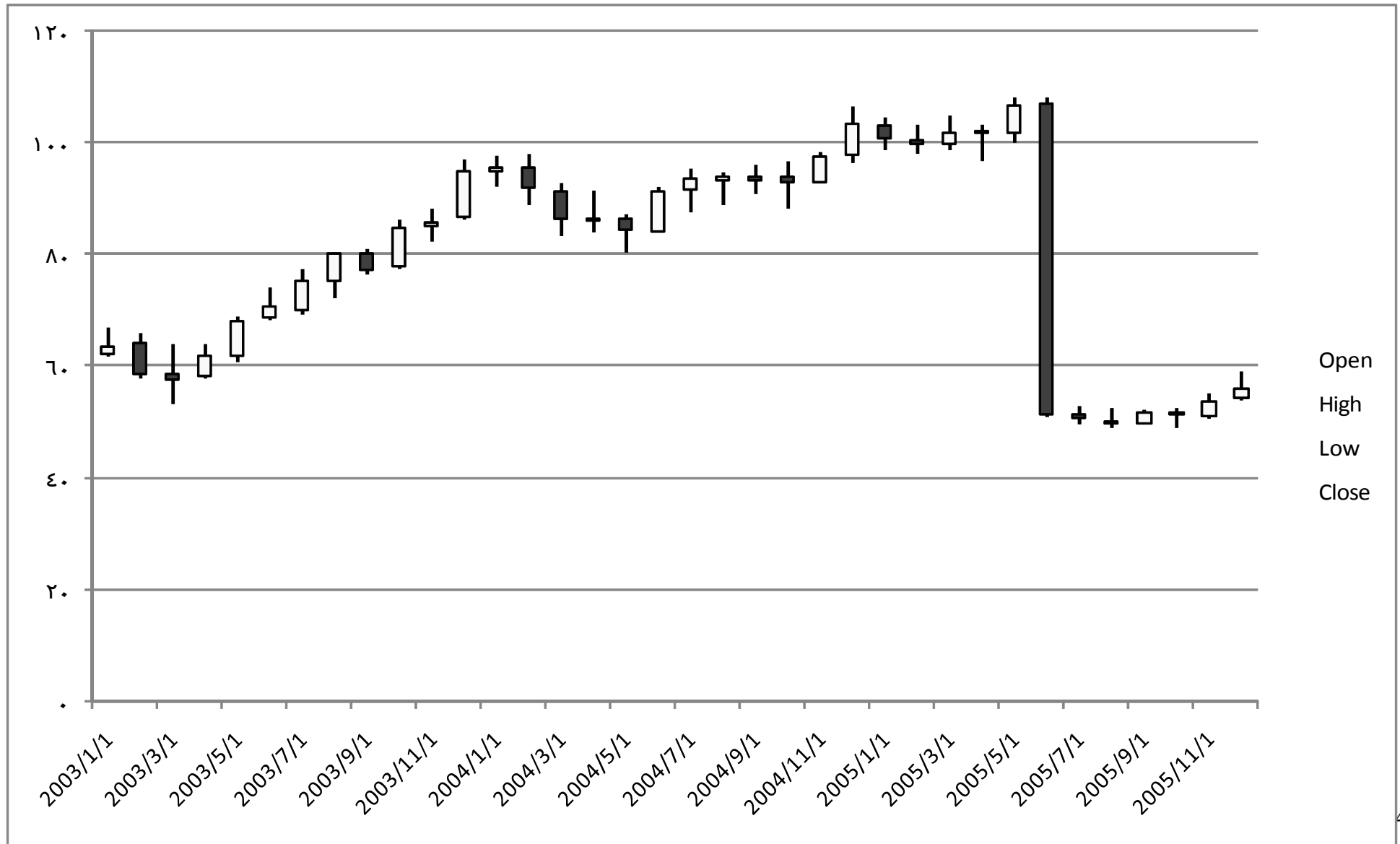


Table 5

Average portfolio returns of CGA on the multiple securities problem

Testing period	Run1	Run2	Run3	Run4	Run5	Average
2003/01/01–03/31	−0.03	−0.26	−0.72	0.24	−1.60	−0.47
2003/04/01–06/30	−1.72	−0.47	−2.87	−1.31	−0.86	−1.45
2003/07/01–09/30	2.31	5.56	2.78	3.5	3.49	3.53
2003/10/01–12/31	3.42	4.12	5.72	4.49	5.22	4.59
2004/01/01–03/31	−1.96	−1.81	−1.72	−2.07	−2.03	−1.92
2004/04/01–06/30	−1.46	−0.27	−2.8	−0.53	−1.32	−1.28
2004/07/01–09/30	−0.65	0.32	0.14	0.37	1.21	0.28
2004/10/01–12/31	5.7	1.61	4.21	3.25	5.03	3.96
2005/01/01–03/31	0.69	1.21	0.33	−1.16	−0.56	0.1
2005/04/01–06/30	4.17	2.13	2.99	2.02	1.46	2.55
2005/07/01–09/30	3.08	2.38	5.01	5.92	4.19	4.12
2005/10/01–12/31	2.7	1.51	2.02	0.73	2.14	1.82
Accumulated	17.11	16.99	15.65	16.18	17.22	16.68

Table 6

Comparisons between CGA and uniform allocation on the multiple securities problem

Testing period	CGA	Uniform
2003/01/01–03/31	−0.47	−0.58
2003/04/01–06/30	−1.45	−1.4
2003/07/01–09/30	3.53	3.27
2003/10/01–12/31	4.59	5.55
2003	6.21	6.85
2004/01/01–03/31	−1.92	−2.45
2004/04/01–06/30	−1.28	−1.35
2004/07/01–09/30	0.28	−0.45
2004/10/01–12/31	3.96	3.15
2004	0.94	−1.18
2005/01/01–03/31	0.10	−0.02
2005/04/01–06/30	2.55	1.9
2005/07/01–09/30	4.12	2.8
2005/10/01–12/31	1.82	1.34
2005	8.83	6.14
Accumulated	16.68	12.06

Thank you!!